LECTURE NOTES

Financial Projections & Requirements

Key Terminologies | Simple Calculations | Practical Examples

Reference: Business Plan Development Framework | Entrepreneurship — Barringer & Ireland

Overview for the Professor

This document covers **10 core financial terminologies** extracted from the Business Plan Development Framework. Each term is defined in plain language, supported by a simple numerical example and formula — ready for classroom delivery without prior finance background required from students.

01 Revenue Forecasting

Revenue Forecasting

What It Means (Simple Definition):

Revenue forecasting is the process of estimating how much money a business will earn over the next 3–5 years. It answers the question: How much will we sell, and at what price?

The Two Main Approaches:

BOTTOM-UP APPROACH	TOP-DOWN APPROACH
Start small — build up from real numbers	*Start big — take a slice of the total market*
Count expected customers, multiply by price and conversion rate	Total Market Size x Your Expected Market Share %

FORMULA — Bottom-Up Revenue Forecast

Revenue = Number of Customers × Average Revenue per Customer × Conversion Rate

Example: 500 potential customers × PKR 5,000 avg. order × 20% conversion = PKR 500,000 monthly revenue

Simple Classroom Scenario:

Year	Customers (Projected)	Revenue @ PKR 5,000/customer
Year 1	100	PKR 500,000
Year 2	250	PKR 1,250,000

Year 3	500	PKR 2,500,000

Conservative (Year 1), Base (Year 2), Upside (Year 3) — This range of scenarios is called Sensitivity Analysis (explained below).

02 Cost Structure & Expense Projections

Cost Structure

What It Means:

Cost structure describes the types and proportions of expenses a business incurs. Costs are divided into two categories — Fixed and Variable.

FIXED COSTS

Stay the same regardless of sales volume

- Rent / Office Lease: PKR 50,000/month
- Salaries (permanent staff): PKR 150,000/month
- Insurance: PKR 10,000/month
- Internet & Utilities: PKR 8,000/month

VARIABLE COSTS

Change directly with how much you produce or sell

- Raw materials: PKR 200 per unit sold
- Packaging & shipping: PKR 50 per order
- Sales commissions: 5% of each sale
- Customer acquisition cost: PKR 300/customer

FORMULA — Total Monthly Cost

Total Cost = Fixed Costs + (Variable Cost per Unit × Units Produced)

Example: PKR 218,000 (fixed) + (PKR 250 × 400 units) = PKR 218,000 + PKR 100,000 = PKR 318,000 total

03 Break-Even Analysis

Break-Even Analysis

What It Means:

Break-even is the point where your total revenue exactly equals your total costs — you are making neither a profit nor a loss. It tells you: How many units do I need to sell just to cover my costs?

FORMULA — Break-Even Point (Units)

Break-Even Units = Fixed Costs ÷ (Selling Price per Unit − Variable Cost per Unit)
The bottom part (Selling Price − Variable Cost) is called the Contribution Margin per Unit

Step-by-Step Example:

Item	Value	Description
Selling Price per Unit	PKR 1,000	What customer pays
Variable Cost per Unit	PKR 400	Cost to make one unit
Contribution Margin	PKR 600	PKR 1,000 − PKR 400
Total Fixed Costs/Month	PKR 120,000	Rent + salaries etc.
Break-Even Units	200 units	PKR 120,000 ÷ PKR 600
Break-Even Revenue	PKR 200,000	200 units × PKR 1,000

NOTE *Selling fewer than 200 units = LOSS. Selling more than 200 units = PROFIT. The lower this number, the less risky the business.*

04 Pro Forma Financial Statements

Pro Forma Financial Statements

What It Means:

Pro forma simply means projected or estimated in advance. These are financial statements prepared for the future — before the business actually operates — to show investors and lenders what the finances might look like.

Three Key Pro Forma Statements:

Statement	What It Shows	Simple Analogy
Income Statement	Revenue − Expenses = Profit/Loss	Your salary minus bills = savings or debt
Cash Flow Statement	Cash coming IN vs. going OUT each month	Your bank account month by month
Balance Sheet	What you OWN (assets) vs. what you OWE (liabilities)	Net worth snapshot at one point in time

Simple Pro Forma Income Statement Example (Year 1):

ITEM	AMOUNT (PKR)
Total Revenue	2,400,000
Less: Cost of Goods Sold (COGS)	(960,000)
Gross Profit	**1,440,000**
Less: Operating Expenses (Salaries, Rent, Marketing)	(900,000)
Operating Profit (EBIT)	**540,000**
Less: Taxes (30%)	(162,000)
NET PROFIT (After Tax)	**378,000**

05 Cash Flow Statement & Working Capital

Cash Flow Statement

What It Means:

A business can be profitable on paper but still run out of cash. Cash flow tracks real money moving in and out each month. Positive cash flow = more money came in than went out.

FORMULA — Net Cash Flow

Net Cash Flow = Total Cash Inflows − Total Cash Outflows

Example: PKR 300,000 received from customers − PKR 250,000 paid to suppliers & staff = +PKR 50,000 (positive)

Monthly Cash Flow Tracker (Simple Example):

Month	Cash IN (PKR)	Cash OUT (PKR) | Net (PKR)
January	150,000	200,000 | -50,000 ⚠ DEFICIT
February	220,000	200,000 | +20,000 ✓
March	310,000	200,000 | +110,000 ✓

NOTE *A business showing profit in the Income Statement can still FAIL because of negative cash flow (e.g., customers have not paid yet). This is called a cash flow gap — working capital management solves this.*

06 Gross Profit Margin & Net Profit Margin

Profit Margins

What It Means:

Margins tell you what percentage of revenue you actually keep after costs. They are essential for comparing performance across businesses of different sizes.

FORMULA — Gross Profit Margin

Gross Profit Margin (%) = (Revenue − COGS) ÷ Revenue × 100

Example: (PKR 2,400,000 − PKR 960,000) ÷ PKR 2,400,000 × 100 = 60% Gross Margin

FORMULA — Net Profit Margin

Net Profit Margin (%) = Net Profit After Tax ÷ Revenue × 100

Example: PKR 378,000 ÷ PKR 2,400,000 × 100 = 15.75% Net Margin

NOTE *Rule of Thumb: Higher margins = healthier business. A 60% gross margin with a 15% net margin is considered good for a startup. Investors benchmark these against industry averages.*

07 Capital Requirements & Use of Proceeds

Capital Requirements

What It Means:

Capital requirements tell investors exactly how much money the business needs to raise, at which stage (Seed / Series A / Series B), and precisely what it will be spent on. Transparency here builds investor trust.

Funding Stages Explained:

Stage	Purpose	Typical Amount (PKR)
Seed Funding	Prove the idea works — build MVP, hire first team	5M – 20M
Series A	Grow the product and customer base — hire, market	50M – 200M

Series B	Scale nationally or internationally	200M – 1B+

Sample Use of Proceeds Table (Seed Round: PKR 10,000,000):

USE OF FUNDS	AMOUNT (PKR)	% OF TOTAL
Product Development / Technology	3,500,000	35%
Marketing & Customer Acquisition	2,500,000	25%
Team Hiring (6 months salaries)	2,000,000	20%
Working Capital Reserve	1,500,000	15%
Legal, Admin & Miscellaneous	500,000	5%
TOTAL	**10,000,000**	**100%**

08 Valuation & Investor Returns (IRR & Return Multiples)

Valuation & Investment Returns

What It Means:

Valuation is the estimated worth of the business. Investors want to know: If I invest today, how much will I get back and when? Two key metrics are Return Multiple and IRR (Internal Rate of Return).

FORMULA — Return Multiple (Money-on-Money)

Return Multiple = Exit Value of Investment ÷ Amount Invested

Example: Investor puts in PKR 5,000,000. Company sold for PKR 40,000,000. Investor's share = PKR 20,000,000. Multiple = PKR 20M ÷ PKR 5M = 4x Return

Valuation Methods at a Glance:

Method	How It Works	Best Used When
Comparable Company Analysis	Compare to similar businesses recently sold or listed	Mature industry with public data available
Discounted Cash Flow (DCF)	Estimate future cash flows and discount back to today's value	Stable, predictable revenue streams
VC Method	Work backwards from a target exit valuation to determine today's value	Early-stage startups with uncertain cash flows

NOTE *IRR = the annualised percentage return an investor earns. A 40–60% IRR is typically expected in early-stage venture capital. Lower risk investments like real estate target 15–20% IRR.*

09 Sensitivity Analysis (Scenario Planning)

Sensitivity Analysis

What It Means:

Sensitivity analysis tests how the financial outcome changes when one key assumption changes. It answers: What is the best case, base case, and worst case? It shows investors you have thought about risk honestly.

Example: Impact of Changing the Selling Price on Annual Profit

SCENARIO	Selling Price/Unit	Annual Revenue	Net Profit
Worst Case	PKR 800	PKR 1,920,000	PKR 60,000
Base Case	PKR 1,000	PKR 2,400,000	PKR 378,000
Best Case	PKR 1,200	PKR 2,880,000	PKR 738,000

NOTE *Key variables to test in sensitivity analysis: Price, Customer Acquisition Cost (CAC), Market Size, Conversion Rate. Identifying which variable has the most impact tells you where to focus management attention.*

10 Financial Assumptions & Transparency

Financial Assumptions

What It Means:

Financial projections are only as credible as the assumptions behind them. Assumptions are the explicit starting beliefs on which all calculations rest — e.g., market growth rate, price per unit, number of customers. Making assumptions transparent invites scrutiny and builds credibility.

Sample Assumption Log for a Business Plan:

ASSUMPTION	VALUE USED	BASIS / JUSTIFICATION
Monthly website visitors	**5,000**	*Based on Google Ads estimate for target keyword volume*

Conversion rate (visitor → buyer)	**2%**	*Industry benchmark for e-commerce startups*
Average order value	**PKR 1,000**	*Derived from competitive pricing analysis*
Customer churn rate per month	**5%**	*Conservative estimate based on SaaS benchmarks*
Annual market growth rate	**15%**	*PBS e-commerce sector growth report 2023*
Cost of goods sold (COGS %)	**40% of revenue**	*Based on supplier quotes obtained in Month 1*

NOTE *False precision (e.g., claiming exactly 2,347 customers in Year 1) is a red flag to investors. Transparent, range-based assumptions with cited sources demonstrate analytical rigor and intellectual honesty.*

PROFESSOR'S QUICK REFERENCE — All 10 Terms & Formulas

#	TERM	CORE FORMULA	KEY TAKEAWAY
01	**Revenue Forecasting**	*Customers × Avg. Revenue × Conversion Rate*	Project income across 3–5 years with 3 scenarios
02	**Cost Structure**	*Total Cost = Fixed + (Variable × Units)*	Classify every expense as fixed or variable
03	**Break-Even Analysis**	*Fixed Costs ÷ (Price − Variable Cost/Unit)*	Minimum units to sell before making profit
04	**Pro Forma Statements**	*Revenue − Expenses = Net Profit*	Three statements: Income, Cash Flow, Balance Sheet
05	**Cash Flow**	*Net Cash Flow = Cash In − Cash Out*	Profit ≠ Cash; track monthly liquidity
06	**Profit Margins**	*Net Margin = Net Profit ÷ Revenue × 100*	% of each rupee of sales you keep
07	**Capital Requirements**	*Seed / Series A / Series B stages*	Detail exactly what investor money will be used for
08	**Valuation & IRR**	*Return Multiple = Exit Value ÷ Amount Invested*	Show 40–60% IRR for VC-stage startups
09	**Sensitivity Analysis**	*Test: Best / Base / Worst Case scenarios*	Prove you understand your key risk variables
10	**Financial Assumptions**	*Assumption → Value → Justification log*	Cite every assumption; avoid false precision

Reference: Barringer, B. R., & Ireland, R. D. (2019). Entrepreneurship: Successfully Launching New Ventures (6th ed.). Pearson.